

REPORT

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Topic: CNN for different batch size, learning rate and epochs.

Colab Link:

<https://colab.research.google.com/drive/1NLYmXrofGD9TDMw6ZVNkJJjJBZwrIjvo?usp=sharing>

CNN:

LOW SETTINGS

"batch_size": 16

"learning_rate": 1e-5

"epochs": 5

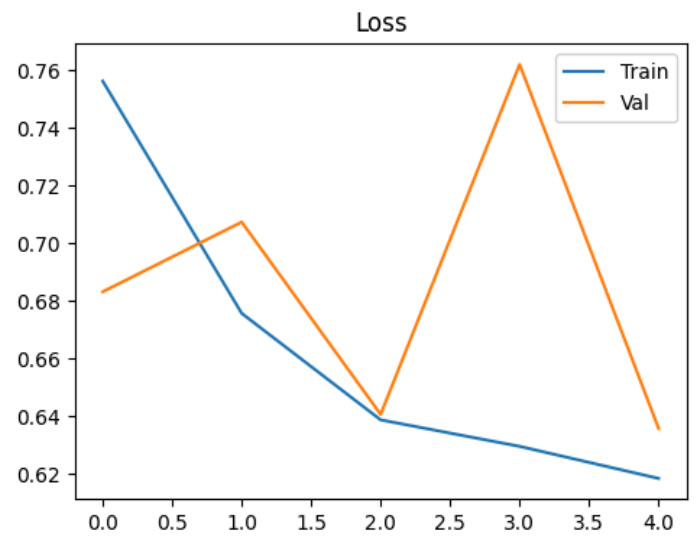
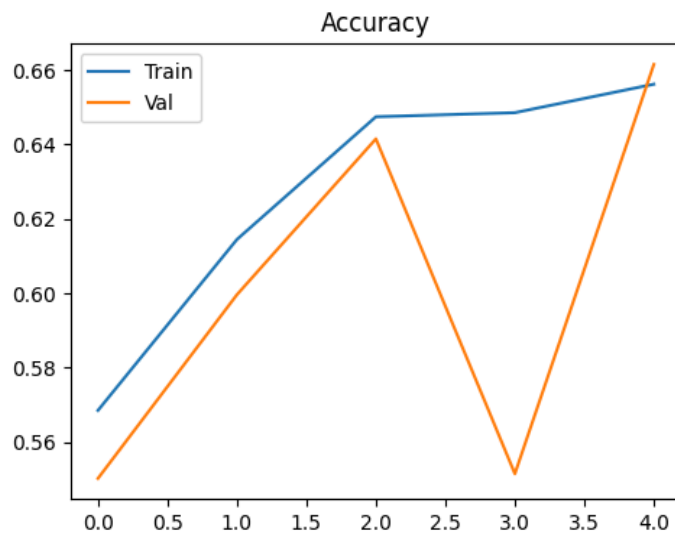
LOW - BEST MODEL & TRAINING LOG

 BEST MODEL: Val Loss: 0.6357, Val Acc: 66.15%

 TRAINING LOG SUMMARY

Epoch 01		Loss: 0.7561		Acc: 56.86%		Val Loss: 0.6831		Val Acc: 55.03%
Epoch 02		Loss: 0.6756		Acc: 61.45%		Val Loss: 0.7072		Val Acc: 59.96%
Epoch 03		Loss: 0.6386		Acc: 64.74%		Val Loss: 0.6405		Val Acc: 64.15%
Epoch 04		Loss: 0.6294		Acc: 64.85%		Val Loss: 0.7619		Val Acc: 55.15%
Epoch 05		Loss: 0.6183		Acc: 65.62%		Val Loss: 0.6357		Val Acc: 66.15%

LOW - ACCURACY AND LOSS GRAPH



LOW - TRUE VS PREDICTIONS

T: dogs
P: cats



T: cats
P: cats



T: dogs
P: dogs



T: dogs
P: dogs



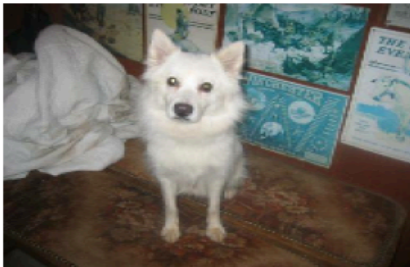
T: cats
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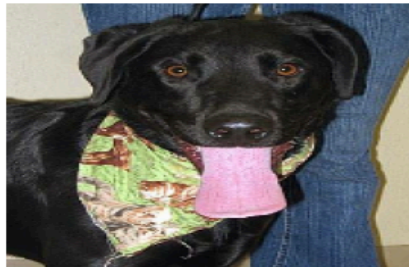
T: cats
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MEDIUM SETTINGS

"batch_size": 32

"learning_rate": 2e-4

"epochs": 30 (EarlyStopping will prevent waste)

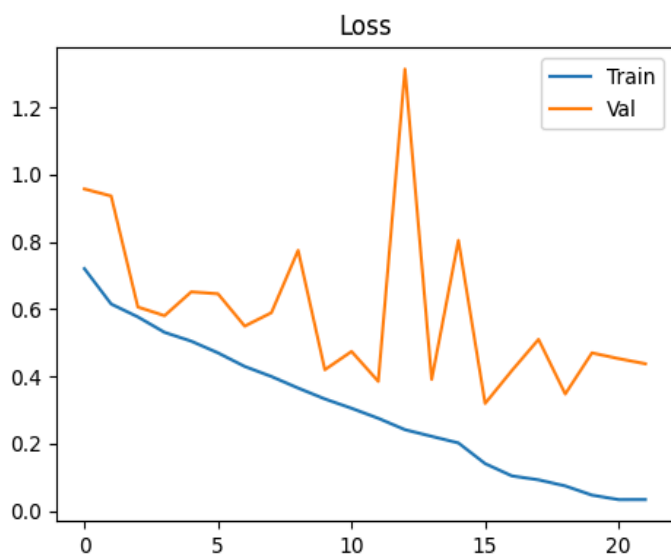
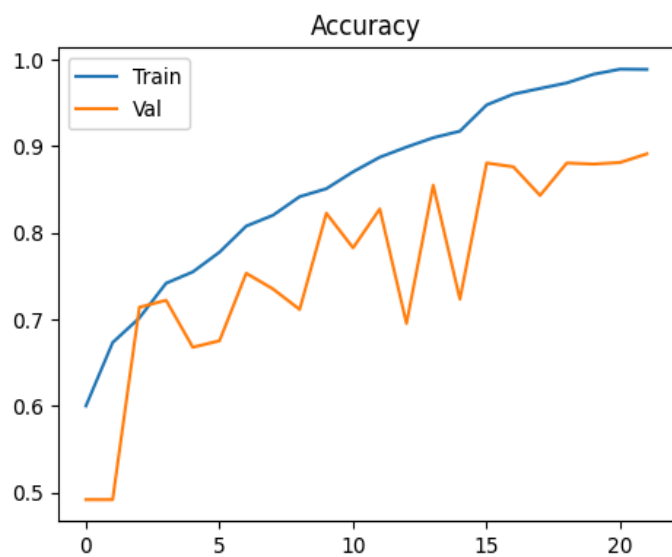
MEDIUM - BEST MODEL & TRAINING LOG

🏆 BEST MODEL: Val Loss: 0.3193, Val Acc: 88.07%

📊 TRAINING LOG SUMMARY

Epoch 01	Loss: 0.7204	Acc: 59.99%	Val Loss: 0.9575	Val Acc: 49.16%
Epoch 02	Loss: 0.6152	Acc: 67.32%	Val Loss: 0.9366	Val Acc: 49.16%
Epoch 03	Loss: 0.5768	Acc: 70.16%	Val Loss: 0.6063	Val Acc: 71.39%
Epoch 04	Loss: 0.5308	Acc: 74.17%	Val Loss: 0.5805	Val Acc: 72.20%
Epoch 05	Loss: 0.5046	Acc: 75.48%	Val Loss: 0.6514	Val Acc: 66.77%
Epoch 06	Loss: 0.4698	Acc: 77.76%	Val Loss: 0.6456	Val Acc: 67.52%
Epoch 07	Loss: 0.4294	Acc: 80.78%	Val Loss: 0.5494	Val Acc: 75.33%
Epoch 08	Loss: 0.3995	Acc: 82.03%	Val Loss: 0.5893	Val Acc: 73.52%
Epoch 09	Loss: 0.3651	Acc: 84.18%	Val Loss: 0.7749	Val Acc: 71.14%
Epoch 10	Loss: 0.3327	Acc: 85.10%	Val Loss: 0.4197	Val Acc: 82.26%
Epoch 11	Loss: 0.3050	Acc: 87.05%	Val Loss: 0.4740	Val Acc: 78.26%
Epoch 12	Loss: 0.2751	Acc: 88.74%	Val Loss: 0.3851	Val Acc: 82.76%
Epoch 13	Loss: 0.2412	Acc: 89.91%	Val Loss: 1.3141	Val Acc: 69.52%
Epoch 14	Loss: 0.2214	Acc: 90.99%	Val Loss: 0.3909	Val Acc: 85.51%
Epoch 15	Loss: 0.2018	Acc: 91.76%	Val Loss: 0.8047	Val Acc: 72.33%
Epoch 16	Loss: 0.1403	Acc: 94.78%	Val Loss: 0.3193	Val Acc: 88.07%
Epoch 17	Loss: 0.1037	Acc: 96.03%	Val Loss: 0.4170	Val Acc: 87.63%
Epoch 18	Loss: 0.0920	Acc: 96.69%	Val Loss: 0.5099	Val Acc: 84.32%
Epoch 19	Loss: 0.0738	Acc: 97.33%	Val Loss: 0.3475	Val Acc: 88.07%
Epoch 20	Loss: 0.0465	Acc: 98.33%	Val Loss: 0.4697	Val Acc: 87.95%
Epoch 21	Loss: 0.0331	Acc: 98.92%	Val Loss: 0.4527	Val Acc: 88.13%
Epoch 22	Loss: 0.0332	Acc: 98.89%	Val Loss: 0.4374	Val Acc: 89.13%

MEDIUM - ACCURACY AND LOSS GRAPH

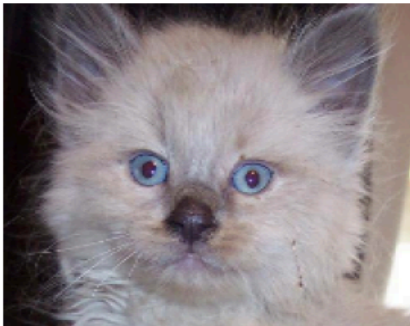


MEDIUM - TRUE VS PREDICTIONS

T: cats
P: cats



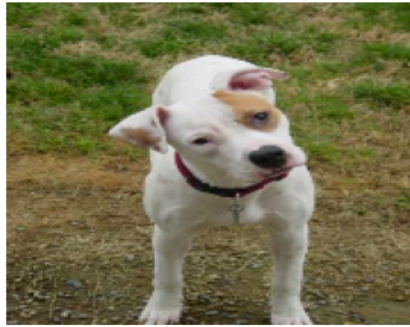
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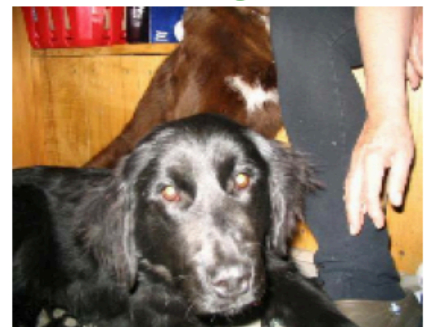
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HIGH SETTINGS

"batch_size": 64

"learning_rate": 1e-3

"epochs": 100 (EarlyStopping will prevent waste)

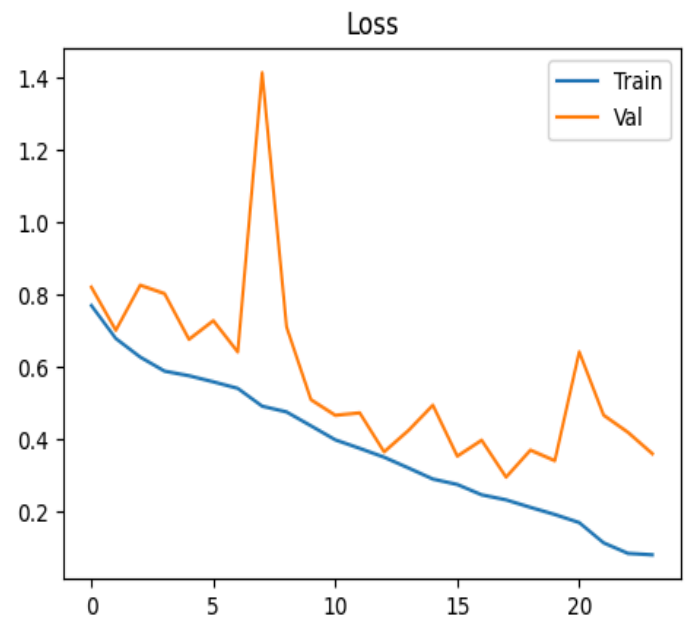
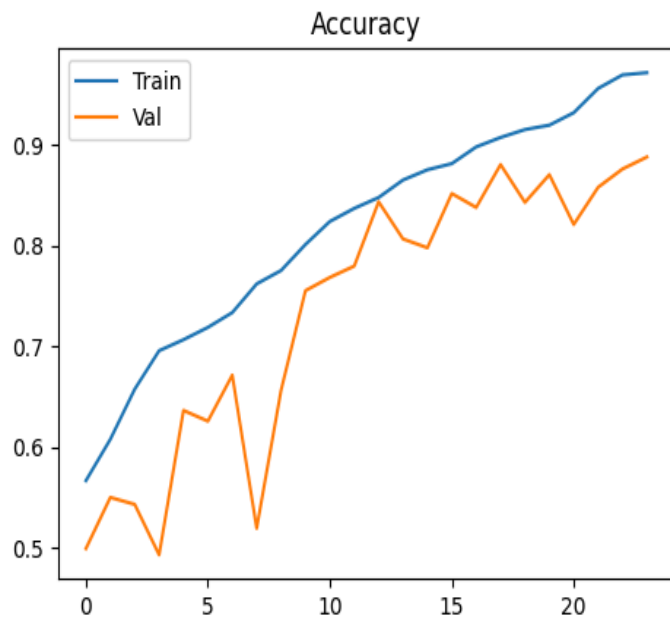
HIGH - BEST MODEL & TRAINING LOG

🏆 BEST MODEL: Val Loss: 0.2951, Val Acc: 88.01%

📊 TRAINING LOG SUMMARY

Epoch 01	Loss: 0.7695	Acc: 56.70%	Val Loss: 0.8203	Val Acc: 49.97%
Epoch 02	Loss: 0.6784	Acc: 60.79%	Val Loss: 0.7012	Val Acc: 55.03%
Epoch 03	Loss: 0.6271	Acc: 65.76%	Val Loss: 0.8258	Val Acc: 54.34%
Epoch 04	Loss: 0.5882	Acc: 69.57%	Val Loss: 0.8025	Val Acc: 49.34%
Epoch 05	Loss: 0.5755	Acc: 70.66%	Val Loss: 0.6761	Val Acc: 63.65%
Epoch 06	Loss: 0.5586	Acc: 71.88%	Val Loss: 0.7277	Val Acc: 62.59%
Epoch 07	Loss: 0.5405	Acc: 73.34%	Val Loss: 0.6413	Val Acc: 67.15%
Epoch 08	Loss: 0.4913	Acc: 76.19%	Val Loss: 1.4136	Val Acc: 51.97%
Epoch 09	Loss: 0.4754	Acc: 77.51%	Val Loss: 0.7105	Val Acc: 65.65%
Epoch 10	Loss: 0.4370	Acc: 80.09%	Val Loss: 0.5093	Val Acc: 75.52%
Epoch 11	Loss: 0.3981	Acc: 82.37%	Val Loss: 0.4661	Val Acc: 76.83%
Epoch 12	Loss: 0.3746	Acc: 83.67%	Val Loss: 0.4728	Val Acc: 77.95%
Epoch 13	Loss: 0.3502	Acc: 84.74%	Val Loss: 0.3649	Val Acc: 84.32%
Epoch 14	Loss: 0.3203	Acc: 86.48%	Val Loss: 0.4243	Val Acc: 80.64%
Epoch 15	Loss: 0.2898	Acc: 87.49%	Val Loss: 0.4940	Val Acc: 79.76%
Epoch 16	Loss: 0.2747	Acc: 88.10%	Val Loss: 0.3530	Val Acc: 85.13%
Epoch 17	Loss: 0.2461	Acc: 89.76%	Val Loss: 0.3971	Val Acc: 83.76%
Epoch 18	Loss: 0.2322	Acc: 90.69%	Val Loss: 0.2951	Val Acc: 88.01%
Epoch 19	Loss: 0.2115	Acc: 91.47%	Val Loss: 0.3696	Val Acc: 84.26%
Epoch 20	Loss: 0.1917	Acc: 91.91%	Val Loss: 0.3406	Val Acc: 87.01%
Epoch 21	Loss: 0.1693	Acc: 93.14%	Val Loss: 0.6415	Val Acc: 82.07%
Epoch 22	Loss: 0.1133	Acc: 95.55%	Val Loss: 0.4664	Val Acc: 85.76%
Epoch 23	Loss: 0.0842	Acc: 96.89%	Val Loss: 0.4189	Val Acc: 87.57%
Epoch 24	Loss: 0.0802	Acc: 97.11%	Val Loss: 0.3598	Val Acc: 88.76%

HIGH - ACCURACY AND LOSS GRAPH



HIGH - TRUE VS PREDICTIONS

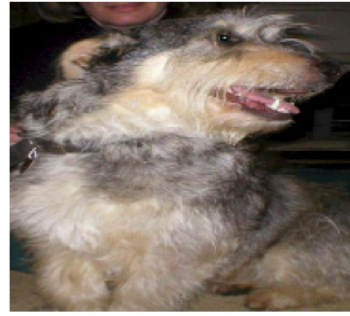
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CONCLUSION

Configuration	Batch Size	Learning Rate	Epochs	Final Val. Accuracy	Key Observation

Low	16	1×10^{-5}	5	66.15%	Underfitted: Learning rate was too slow for only 5 epochs.
Medium	32	2×10^{-4}	19*	88.26%	Optimal: Best balance of speed and generalization.
High	64	1×10^{-3}	18*	88.01%	Volatile: High accuracy but showed signs of validation loss instability.